

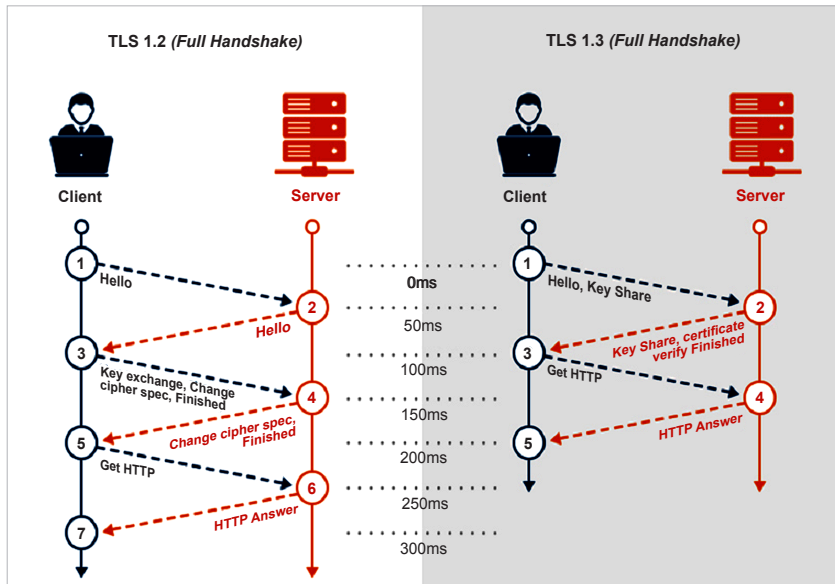


Figure 11: The advantages of TLS v1.3

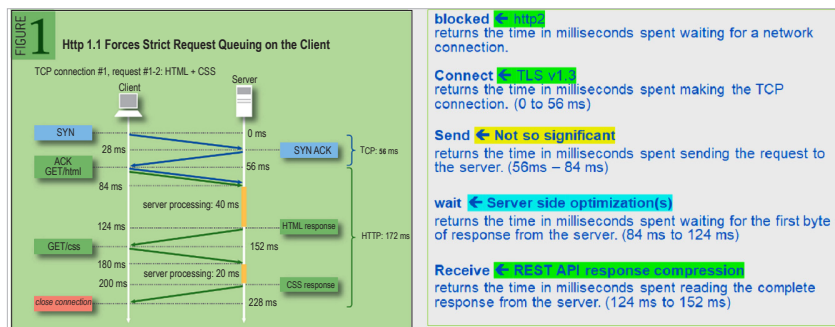


Figure 12: Putting everything together

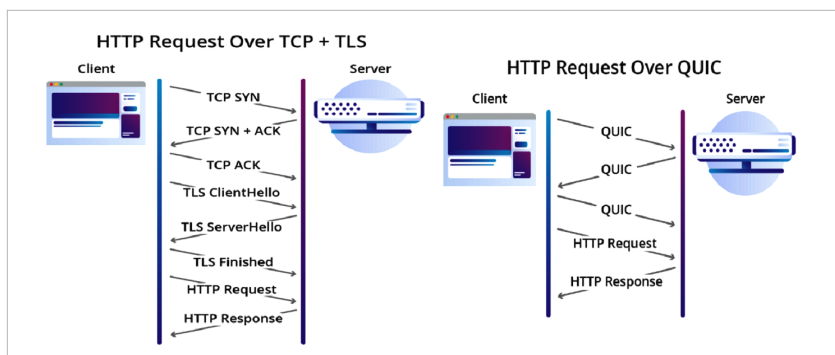


Figure 13: Http3 (aka QUIC) protocol stack

| Scenario                    | Elapsed time | Receive time | Data transfer |
|-----------------------------|--------------|--------------|---------------|
| Http1.1 without compression | 10s          | 7s           | 10MB          |
| Http1.1 with compression    | 5s           | 2s           | 2.6MB         |
| Http2 without compression   | 9.7s         | 6.4s         | 10MB          |
| Http2 with compression      | 2.5s         | 1s           | 2.6MB         |

Table 1: REST API compression results

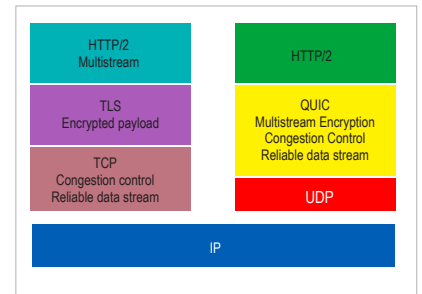


Figure 14: The QUIC protocol architecture

following setup:

1. We placed the client at Bengaluru and the server on the East Coast, USA (North Carolina).
2. We saw that roughly 1.3MB of data was used for 1000 records. And some of the AutoSupport was used for 8000 records max. So, roughly, 10MB data size was taken as representation in this experiment.
3. API returns 30k records, compression level=6 (default), chunkSize=16384 bytes, threshold=0.

And the results are stunning (in our experimental setup), as shown in Table 1.

One major advantage of putting all these together is that there is nothing to be changed in the client-side code, as all the code is at the server side only. Browser support for http2 is available. Figure 12 depicts the strategy one can adopt for putting all the above features together in the same basket.

So what lies ahead? Well, there is a lot happening that is very promising. Let us look at it one-by-one.

## Http3

The new kid on the block is QUIC or http3. Google and Facebook are already latched on to it. It is UDP driven at layer-4 (transport).

The advantages of Http3 over its predecessors are:

1. Till now the OS (kernel) abstracted out the network stack implementation. So making changes was difficult, and it was time consuming to try and test these changes.

*Continued on page 54...*